PCDP – Post-Coding Development Paradigm

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Section 1

What

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Human Intent, Machine Implementation

Domain experts write **specifications**.
AI generates **all** implementation code.
Engineers never write implementation code directly.

PCDP

Post-Coding Development Paradigm

Write Specs, Not Code

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This is not “AI-assisted coding”

	Traditional	Vibe Coding	**PCDP**
Human writes	code	code + prompts	**specs only**
AI role	none	suggests	**translates**
Primary artifact	source code	source code	**specification**
Target language	developer	developer	**template**
Regulated domains	manual audit	prohibited	**enabled**

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The specification

```
## BEHAVIOR: transfer
INPUTS:
  from: Account
  amount: Amount
STEPS:
  1. Validate preconditions.
  2. Debit from.balance.
  3. Credit to.balance.
POSTCONDITIONS:
  - SUM(balances) unchanged
EXAMPLES:
  EXAMPLE: success
  GIVEN: balance = 100
  WHEN:  transfer(30)
  THEN:  balance = 70
```

No programming language.
No target platform.
No toolchain knowledge required.
The **template** decides all of that.

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Section 2

Why

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1. Separation of concerns

The spec says **what** and **where**.
The template decides **how**:

- programming language
- packaging format
- toolchain conventions
- deployment target

Change language across the whole project?
Change one preset file.
The spec is untouched.
A spec written today is valid in 2045 —
regardless of what language is in fashion.

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2. Long-term maintainability

Specs are more stable than code.
Code accumulates technical debt.
Specs describe intent — intent rarely changes.

When requirements change:

- update the spec
- regenerate the code
- no manual refactoring

The spec is the documentation.
Not a comment that drifts from the code.
Not a wiki page nobody updates.
The running system is always derived from the spec.

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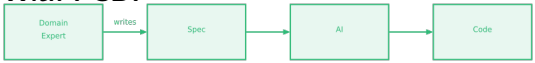
3. Domain experts author directly

Today

Every handoff loses information.
Requirements get simplified.
Misunderstandings go undetected.

With PCDP



The cardiologist specifies
the device behaviour directly.
No interpreter in the middle.

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And: formal verification is available

The LLM translation is probabilistic — we do not claim otherwise.

For components that **require** stronger guarantees:

Lean 4, F*, or Dafny can be added as an **optional verification layer** — without changing the specification format.
The spec stays the same. The verification path is a template choice.

- memory safety by construction
- formal proofs of invariants
- state machine correctness
- ISO 26262 / DO-178C evidence

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Proof: pcdp-lint

`pcdp-lint` — the specification validator — was **specified and generated using PCDP itself.**

Zero hand-written implementation code.

Tested across multiple AI models, three continents, one result:

Every model resolved Go from the template
without being told.

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Section 3

How

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The workflow

1. Domain expert writes a spec (or AI interviews them)
2. `pcdp-lint` validates structure
3. Deployment template resolves the language
4. LLM translates spec → code + audit bundle

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Language is never your problem

The spec declares **what** and **where** (deployment context).

The template resolves **language, packaging, conventions**.

A spec written today is valid if the organisation
switches from Go to Rust in 2045 — **no spec change needed**.

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The audit bundle

Every translation produces:

- specification (human-reviewable)
- generated source code
- packaging artifacts (RPM, DEB, OCI)
- independent test suite
- translation report
- workflow diagram (Pikchr)
- metadata + hashes

Designed for:

- ISO 26262 (automotive)
- DO-178C (aviation)
- IEC 62304 (medical)
- Common Criteria EAL4+
- EU Cyber Resilience Act

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Getting started

Write a spec (no format knowledge needed):

```
# AI interviews the domain expert:
ollama run llama3.2 \
  "$(cat prompts/interview-prompt.md)"
```

Then validate:

```
pcdp-lint myspec.md
```

Translate to code:

```
# mcphost with the translator prompt
# reads template + spec,
# produces code + audit bundle
```

Everything in the repo:
github.com/mge1512/pcdp

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Section 4

The specification

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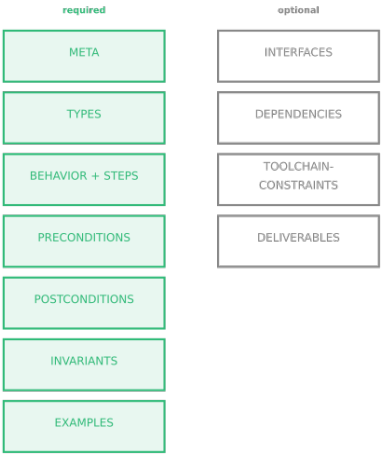
Spec sections



`pcdp-lint` validates structure before any translation begins.

Key constraints:

- BEHAVIOR must include `STEPS:`
- INVARIANTS carry `[observable]` or `[implementation]` tags
- EXAMPLES must cover error paths
- No programming language in spec

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BEHAVIOR in depth

```
## BEHAVIOR: shutdown
Constraint: required
INPUTS:
  vm: VirtualMachine
PRECONDITIONS:
  - vm.state = Running
STEPS:
  1. Send shutdown signal.
  2. Record shutdown-start time.
     MECHANISM: do NOT block;
     requeue immediately.
  3. On re-entry: check elapsed.
     If > timeout: Destroy(vm).
POSTCONDITIONS:
  - vm.state = Stopped
```

`STEPS` = the algorithm, not just the contract.

`MECHANISM:` = where the *how* matters for correctness.

`Constraint:` field:

- `required` — always implement
- `supported` — preset activates it
- `forbidden` — never implement

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INTERFACES: testable abstractions

```
## INTERFACES

Transport {
  required-methods:
    Connect(ctx, spec) -> (Session, error)
  implementations-required:
    production:  RealTransport
    test-double: FakeTransport {
      configurable: ConnectErr, Sessions
    }
}
```

Independent tests use **only** declared test doubles.
`go test ./independent_tests/` runs without any live service.

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Section 5

Templates and translation

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Deployment templates

Templates encode **everything** the spec does not:

- target language + alternatives
- binary type, packaging formats
- CLI conventions
- security constraints
- TYPE-BINDINGS
- delivery phases (EXECUTION)
- compile gate

One preset file changes language for the whole project. The spec is untouched.

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TYPE-BINDINGS: no translator discretion

Template excerpt:

```
## TYPE-BINDINGS
| Spec type   | LANGUAGE=Go            |
|-------------|------------------------|
| Duration    | metav1.Duration        |
| Timestamp   | metav1.Time            |
| Condition   | metav1.Condition       |
| List<T>     | []T                    |
```

Without TYPE-BINDINGS: one translator produces `string`, another produces `metav1.Duration` — incompatible at the API level.

With TYPE-BINDINGS: **deterministic, language-specific types** — applied mechanically, not by translator judgement.

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The translation phases

1 Core implementation

2 Build + packaging

3 Test infrastructure

4 Documentation

5 Compile gate

Defined in the template's `## EXECUTION` section — **not** in the prompt.

Resume logic: partial runs restart from the first missing file.

Compile gate is language-specific:

```
go mod tidy    # direct deps only
               # no fabricated versions
go build ./...
```

`TRANSLATION_REPORT.md` is **always last** — its presence signals a complete run.

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Section 6

Tooling

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pcdp-lint

What it validates:

- required sections present
- META fields + SPDX license
- deployment template resolves
- BEHAVIOR has STEPS
- INVARIANTS have tags
- EXAMPLES cover error paths
- Constraint: field values
- cross-section consistency

Self-hosting:

`pcdp-lint` was specified and generated using PCDP itself.

Running `pcdp-lint` against its own spec passes — the paradigm validates itself.

```
pcdp-lint myspec.md
pcdp-lint strict=true myspec.md
pcdp-lint list-templates
```

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Translation confidence

Every `TRANSLATION_REPORT.md` includes:

EXAMPLE	Confidence	Verification method	Unverified claims
vm-start	High	`TestVMStart` — FakeSession	libvirt UUID format
graceful-stop	Medium	`TestGracefulStop` pass 1 only	timeout path
host-ready	Low	none	entire sequence

- **High** = named test in `independent_tests/` passes without live services
- **Medium** = partial test coverage
- **Low** = reasoning only — no test

Unverified claims must be listed **explicitly**. Never silently omitted.

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Large projects

Architect defines interfaces first:

```
## INTERFACE
EXPORTS:
  TYPES:  Account, TransferResult
  BEHAVIOR: transfer
  INVARIANTS:
    - GLOBAL: balance >= 0
```

Component specs declare imports:

```
Imports:
  - account-svc: ./account.md
                 #INTERFACE@>=1.2.0
```

Rules:

- interfaces before implementation
- build order = topological sort
- no circular dependencies
- component fits in one LLM context window

pcdp-lint v2 (planned): validates full project graph — imports, circular deps, interface versioning.

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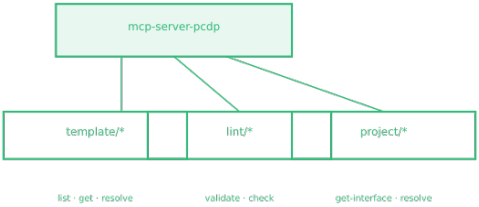
mcp-server-pcdp



Any MCP-capable host connects directly:

- **mcphost** (CLI)
- **VS Code** with Copilot
- **Claude Desktop**
- custom agents

Preset layering server-side:
vendor → `/etc/pcdp-server/` → team → user

The LLM is the wizard.
The server is the data layer.

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